

User's Guide for the Vertical City Weather Generator (VCWG v1.6.2)

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1 About VCWG

The Vertical City Weather Generator (VCWGv1.6.2) framework employed in this study is based on our previously developed and validated model, which simulates urban building energy performance and associated GHG emissions [Aliabadi et al., 2021, Moradi et al., 2021, Moradi, 2021, Aliabadi, 2022, Aliabadi et al., 2023, Safdari et al., 2024, Safdari et al., 2025, Madadizadeh et al., 2025, Madadizadeh et al., 2026]. In the present work, the core modeling structure remains unchanged; however, the framework is extended to incorporate future climate conditions through weather files generated at five-year intervals from 2025 to 2050 under three SSPs (SSP1-2.6, SSP2-4.5, and SSP5-8.5). This extension enables the evaluation of retrofit strategies under evolving temperature patterns, radiation conditions, and seasonal dynamics, which directly influence building energy demand and system performance. By applying the model across multiple Canadian cities representing diverse climate zones (Vancouver (Zone 4), Toronto (Zone 5), Montreal (Zone 6), Halifax (Zone 6), St. John's (Zone 6), Calgary (Zone 7A), Winnipeg (Zone 7A), Saskatoon (Zone 7A), Whitehorse (Zone 7B), and Yellowknife (Zone 8) [ASHRAE, 2020, Madadizadeh et al., 2025]), the study captures how projected climate variability affects the effectiveness of both passive and active retrofit measures. The integration of future weather scenarios provides a forward-looking assessment of building performance, allowing the identification of retrofit strategies that remain robust and effective over time, while also revealing how optimal solutions shift under different climate pathways toward 2050.

VCWG v1.6.2 is shared under the GNU General Public License Version 3. The terms and conditions of the license are accessible via: <https://www.gnu.org/licenses/gpl-3.0.en.html>. Please do not distribute VCWG v1.6.2 to third parties. Instead, please refer interested groups to the At-

atmospheric Innovations Research (AIR) Laboratory to acquire a copy of VCWG v1.6.2. Please consider offering co-authorship to AIR lab members if VCWG v1.6.2 is used significantly toward the completion of a project.

2 Basic Setup

2.1 Setting the Climate Forcing Files

Future weather files were developed to represent projected climate conditions for the period 2025–2050. Climate data were obtained from the Ouranos PAVICS platform, which provides access to high-resolution regional climate model outputs from the CORDEX North America (NAM-12) dataset [Ouranos and CRIM, 2018, World Climate Research Programme, 2021]. Hourly climate variables corresponding to three Shared Socioeconomic Pathways (SSP1-2.6, SSP2-4.5, and SSP5-8.5) were extracted and processed to generate future EnergyPlus weather (*.epw) files for each study city. These EPW files were then used as meteorological inputs for the VCWG simulations by placing them in the corresponding weather directory (e.g., “/resources/epw/”). In the released versions of VCWG, baseline weather files are typically prepared using the ERA5 data product from the European Centre for Medium-Range Weather Forecasts (ECMWF) [Aliabadi and McLeod, 2023], while in this study future climate-adjusted EPW files were developed using CORDEX-derived climate projections from the PAVICS platform.

2.2 Setting Input Parameters

VCWG can take input parameters from the files located in the directory “/resources/parameters/”. These files contain the required parameters of the case study including urban characteristics, vegetation parameters, view factors, simulation parameters, and building renewable and alternative energy configurations. In the released version of the software there are 12 input parameter files for Toronto. To consider the renewable energy options for the model, variable “Adv_ene_heat_mode” should be set. To include only photovoltaic panels and wind turbine, this parameter should be set to 3. For inclusion of other renewable energy options (solar thermal, building integrated thermal energy storage, phase change materials, and energy recovery), this variable should be set to 1 for heating mode and 0 for cooling mode of operation.

Files “initialize_Toronto_1.uwg” to “initialize_Toronto_12.uwg” are used for annual runs of VCWG, and they are set up to run for each month of the year. Lines 230, 231, 232 contain the starting month, starting day, and the duration of the simulation in number of days. For each month, it is desired to start the simulation 3 days before the start of the month and then discard the first 3 days of data as spin up data. For example, to simulate February, one can start from Month = 1 (January), Day = 29 (3 days before start of February), for nDay = 31 (28 days in February plus 3 days for spin up). The January simulation is an exception because the EPW files only contain data for single year. The user is able to change the parameters to define and run a simulation of interest.

There are three more files in which input parameter specifications should be made. 1) To change the building envelop properties (e.g. resistance values) and HVAC equipment specification (e.g. coefficient of performance and thermal efficiency) the file “LocationSummary” in di-

rectories “/resources/DOERefBuildings/BLD1” through “/resources/DOERefBuildings/BLD16” should be modified corresponding to building type. Note that in the released version of the software the appropriate folder is “/resources/DOERefBuildings/BLD6” for mid rise apartment. Also note that only the last column of the spreadsheet is advised to be modified corresponding to custom values. i.e. the users are discouraged to change values in the other columns. 2) To change the ventilation and infiltration rates, the appropriate file “ZoneSummary” in directories “/resources/DOERefBuildings/BLD1” through “/resources/DOERefBuildings/BLD16” should be modified corresponding to building type. Note that in the released version of the software the appropriate folder is specified for the mid rise apartment as “/resources/DOERefBuildings/BLD6”. Two copies of the original files “BLD6_LocationSummaryOriginal” and “BLD6_ZoneSummaryOriginal” are kept in directory “/resources/DOERefBuildings/BLD6”. As will be explained later, values in files “BLD6_LocationSummary” and “BLD6_ZoneSummary” may be changed frequently, so it is a good idea to keep a fresh copy of the original values. Before running a new optimization case:

- Delete files “BLD6_LocationSummary” and “BLD6_ZoneSummary”.
- Copy files “BLD6_LocationSummaryOriginal” and “BLD6_ZoneSummaryOriginal”.
- Rename these to “BLD6_LocationSummary” and “BLD6_ZoneSummary”.

If desired, new view factors can be obtained by running “/UWG/Run_RayTracing.py” and copy and paste the results from file e.g. “/UWG/ViewFactor_Toronto.txt” into the input file e.g. “/resources/parameters/initialize_Toronto_0.uwg” or other initialization files.

Economic input parameters are specified in a few places. Under “resources/Economics”, we have many files that specify various needed inputs. Key files, relevant to Toronto: “carbon_intensity_projection”, “ElecEmissionIntensity”, “ON_NG_Residential_Rates”, “ON_TOU_Electricity_Rates”, and “SocialCarbonCost”. Other inputs are specified in “EconomicGHGAnalysis.py”.

The grid emission intensity is based on the values provided in the “ElecEmissionIntensit” text file. At the start of each simulation, the selected starting year must be specified, and the corresponding row from this file should be used. For instance, since 2020 is used as the starting year, row 8 is selected. The relevant implementation can be found in the corresponding section of the code in “EconomicGHGAnalysis.py”.

```

ElecCO2Saving = (MonthlyDiffElecConsump) * A_building *
(CarbonIntensityProjection.iloc[year-1,1] /100) *\
ElecEmissionIntensity.iloc[8, Region] / 1000

```

3 Running VCWG for Optimization Studies

This version of VCWG is designed to run on Python 2.7.13. This version of Python can be downloaded from <https://www.python.org/downloads/release/python-2713/>. For example, for a 64-bit Windows operating system the installation file will be “python-2.7.13.amd64”. The following packages and versions can be used: numpy 1.14.3, scipy 1.1.0, matplotlib 2.2.2. Note that other packages may also work. “UWG/Run_RayTracing.py” is designed to run on Python

3.6.1. This version of Python can be downloaded from the following link <https://www.python.org/downloads/release/python-361/>. For example for a 64-bit Windows operating system the installation file will be “python-3.6.1-amd64”. The following packages and versions can be used: numpy 1.19.5, scipy 1.1.0, matplotlib 3.1.1. Note that other packages may also work.

To run VCWG for micro-genetic optimization, various other settings are necessary. In the associated file “opt.VariableSpace.py”, the user specifies the range for all variables that may be optimized. In addition the increment for change of each variable must be set. That is to say that the micro-genetic optimizer does not optimize a variable in a continuous scale, but rather in a discrete scale. For example, the lower value for photovoltaic coverage is set to “lPV=0.1” per building footprint area [$\text{m}^2 \text{ m}^{-2}$], the higher value for photovoltaic coverage is set to “hPV=0.6” per building footprint area [$\text{m}^2 \text{ m}^{-2}$], and solutions with increments of 0.1 per building footprint area [$\text{m}^2 \text{ m}^{-2}$] are possible. Next, three options are available for optimization. These are specified by setting variable “ene_mode”. When “ene_mode=3”, up to 9 possible variables can be optimized. This mode is associated with choice of only PV and wind turbine as renewable energy. If a value of 1 is chosen for optimization of a variable, then it is optimized; if zero is chosen, then it is not optimized. When “ene_mode=2”, up to 7 possible variables can be optimized. This mode is associated with choice of no renewable energy, in which case, only basic building variables will be optimized. When “ene_mode=None”, up to 20 possible variables can be optimized. This case is associated with utilization of most possible renewable energy options. In this case the advanced energy mode could be either 1 (for heating) or 0 (for cooling) depending on the month of analysis. Among these 20 variables, five are economic parameters, including loan amount, rebate, loan interest rate, and electricity and fuel consumption taxes.

The micro-genetic optimizer needs to update the file “LocationSummary.csv” with new values of building thermal resistance for roof and walls. This is achieved by overwriting the file “LocationSummary.csv” and “ZoneSummary.csv” after inner and outer iterations of the optimizer. Before the simulation begins, it is desirable to copy and rename the original files “BLD6_LocationSummaryOriginal.csv” into “BLD6_LocationSummary.csv” and “BLD6_ZoneSummaryOriginal.csv” into “BLD6_ZoneSummary.csv”. After few iterations of the code the values inside these files will change. The file “opt_UpdateCSV.py” is responsible for overwriting them.

The EPW file and the initialization files are defined inside the file “VCWG162Optimization.py”. For example, in this distribution of the software the EPW file name is “Toronto_ssp126.2025” and the initialization files range from “initialize_Toronto_1.uwg” to “initialize_Toronto_12.uwg” for each month. The “ene_mode” should be defined for each month in this file, to be just like the “Adv_ene_heat_mode” for each month in the input initialization files. The case name is formatted to include the outer iteration number “OuterIter”, the inner iteration number “InnerIter”, and the individual number in the population “Idv”. As the optimizer progresses, it creates buildings with different attributes, each of which are assigned three numbers for identification (i.e. outer iteration, inner iteration, and individual numbers).

The rest of the settings should be in file “opt_Run.py”. This file has 5 main parts: in part 1, the variable constants are set. The “buildingtype=6” is currently set to mid-rise apartment. For the first VCWG run, i.e. the base case with no alternative energy, “base_ene_mode=2”, which means no renewable energy is considered for the first run. The corresponding energy price inflation

rates should be set for gas and electricity e.g.: “ElecPriceInc = 0.01”, “GasPriceInc = 0.01”. The number of objective functions for saving is specified: “NumberOfObjFunSave = 8”. The optimization weights are given, in this case for Total GHG Emissions Savings, Owner Annual Marginal Cost Savings, and Government Annual Marginal Cost Savings: “Wgt0 = -0.2”, “Wgt1 = -0.6”, and “Wgt2 = -0.2”.

The remaining simulation and optimization settings are defined in the “opt_Run.py” file. This file includes the main user-defined parameters required to run the optimization. The “buildingtype” variable should be consistent with the “BLD” variable in the initialization file; in this study, “buildingtype = 6” represents a mid-rise apartment building. The base-case energy mode is specified as “base_ene_mode = 2”, indicating that the initial VCWG run does not include retrofits or renewable/alternative energy systems. The electricity and gas price inflation rates are then defined using “ElecPriceInc” and “GasPriceInc”, respectively; for example, “ElecPriceInc = 0.01” and “GasPriceInc = 0.01”. The climate scenario is selected through “SSPScenario”, where “126”, “245”, and “585” correspond to SSP1-2.6, SSP2-4.5, and SSP5-8.5, respectively. The simulation year is specified using “Year”, such as “Year = 2025”. The number of objective functions saved during the optimization is set by “NumberOfObjFunSave = 8”. Finally, the optimization weights are assigned to the three objective functions considered in this study: total GHG emissions savings, owner annual marginal cost savings, and government annual marginal cost savings. For example, “Wgt0 = -2/10”, “Wgt1 = -6/10”, and “Wgt2 = -2/10” assign weights of 0.2, 0.6, and 0.2 to these objectives, respectively.

```
# Building type should be consistent with "bld" variable in the initialization file
buildingtype = 6
# set the energy mode for base case that SHOULD NOT include retrofits, renewable/alternati
base_ene_mode = 2

ElecPriceInc = 0.01
GasPriceInc = 0.01

# Shared Socioeconomic Pathways (SSPs): SSP1-2.6, sustainability| SSP2-4.5, middle of the
SSPScenario = 126

#Year
Year = 2025

# The number of the objective functions to save
NumberOfObjFunSave = 8

# Optimiztation weights 0: Total GHG Emissions Savings, 2: Owner Annual Marginal Cost Sav

Wgt0 = -2/10      # Optimization weight for total GHG emissions savings
Wgt1 = -6/10      # Optimization weight for owner's annual cost savings
Wgt2 = -2/10      # Optimization weight for government annual cost savings
```

The following constant variables specify population features of optimization:

```
# Total number of the population in each iteration, it MUST be ODD
n_pop = 5
# The new population involved in each iteration, it MUST be EVEN
n_change = n_pop - 1
# Maximum and minimum number of inner iterations for generating new populations
iMaxInner = 6
iMinInner = 2
# Maximum and minimum number of outer iterations for generating new populations
iMaxOuter = 33
iMinOuter = 35
# Convergence threshold for inner iterations
ConvergenceThresholdInner = 0.03
# Convergence threshold for outer iterations
ConvergenceThresholdOuter = 0.001
```

In part 2, the user runs VCWG once for a base case that should not include retrofits, renewable/alternative energy. The desired variables in each of the 12 initialization files should be set. Also the correct starting building envelop thermal resistance value should be set in the “LocationSummary.csv” file. One way is to recover the correct starting value from the original file “LocationSummaryOriginal.csv”. After the optimization starts, up to 13 variables will be updated and passed to the function “opt_VCWG”.

In part 3, the optimization is set up. There are 3 objective functions: Total GHG Emissions Savings, Owner Annual Marginal Cost Savings, and Government Annual Marginal Cost Savings. The “ene_mode” is set to “none” initially, but it will read from the 12 entries of “Adv_ene_heat_mode” variable from “VCWG162Optimization.py” file.

In part 4, the optimization is executed. The code completes inner and outer loop iterations and monitors the convergence criteria. It breaks out of the inner and outer loops either when maximum number of iterations are reached or when convergence criteria are met. Throughout the iteration process, the results are saved in text files, CSV files, and log files.

Currently in each inner loop iteration the solution or individual attributes (genes) are just mixed, while no new individuals are sampled randomly. Currently the sample individual population is 5. After each inner loop iteration, the best individual, i.e. that with lowest overall objective function value, is kept as the “elite individual”. For the elite individual the genetic code of the individual is not changed for the next inner iteration. The worst individual, i.e. that with highest overall objective function value, is discarded. For example, consider the population below, where each individual has 6 attributes (genes). Each gene holds the value of the variables to be optimized (e.g. “V_bites”, “m_dot_st_f”, “A_ST”, “V_pcm”, “T_melt”, “m_dot_he_st”)

- Individual 1: a1 a2 a3 a4 a5 a6 (lowest objective function, elite)
- Individual 2: b1 b2 b3 b4 b5 b6

- Individual 3: c1 c2 c3 c4 c5 c6
- Individual 4: d1 d2 d3 d4 d5 d6
- Individual 5: e1 e2 e3 e4 e5 e6 (highest objective function, discard)

From the remaining 4 individuals (including the elite individual) four children are created by pairing two parents and mixing their genes. The genes are mixed by taking a sequence of individual attributes and exchanging them at an arbitrary point. For the individual sample above, we obtain

- Individual 1: a1 a2 a3 a4 a5 a6 (elite)
- Individual 2: a1 a2 b3 b4 b5 b6 (mixed genes)
- Individual 3: b1 b2 a3 a4 a5 a6 (mixed genes)
- Individual 4: c1 c2 c3 c4 d5 d6 (mixed genes)
- Individual 5: d1 d2 d3 d4 c5 c6 (mixed genes)

If the maximum number of inner iterations are reached, or if the convergence criteria for the inner iteration is satisfied, then the elite individual is kept, while 4 new individuals are randomly sampled to continue to the next inner iteration. For example, from the above sample, the following sample will be generated.

- Individual 1: a1 a2 a3 a4 a5 a6 (elite)
- Individual 2: f1 f2 f3 f4 f5 f6 (new individual)
- Individual 3: g1 g2 g3 g4 g5 g6 (new individual)
- Individual 4: h1 h2 h3 h4 h5 h6 (new individual)
- Individual 5: i1 i2 i3 i4 i5 i6 (new individual)

This process continues, until maximum number of outer iterations are reached, or the convergence criteria for the outer iteration is satisfied.

In part 5, the results are also saved in python array format using hyper dimensions. The optimization process takes a long time. It is better to run the code on a cluster using a single CPU. To speed up the simulation process, it is better to run simultaneous VCWG computations on multiple CPUs. Each CPU can complete simulation for a given configuration, location, year, etc.

Note: VCWG v1.6.2 is designed to run on a highly parallel computational architecture. It needs 12 CPUs, so each month of the year, run on 1 CPU. If you run it on a laptop or computer with fewer than 12 CPUs available, it will print error messages. Ideally, you need a server with 12 CPUs available.

To run the optimization engine, in a server environment, we recommend using the “screen” com-

mand. Do the following:

```
screen -S Simulation1  
python opt_Run.py
```

Note that if multiple versions of python is installed on a server, “python” will automatically execute python 2 (not 3). The simulations in VCWG v1.6.2 are supposed to be run using python 2. You can press “Ctr + A + D” to exit the “screen” mode. To go back to the screen enter:

```
screen -r
```

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